

CS-233 ML Project – Milestone 2 Report

Gaming & Mental Health: Predicting Addiction Metrics

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1 Introduction

This report presents two Machine Learning methods implemented from scratch on the *Gaming and Mental Health Dataset* (same as Milestone 1: 2,000 users, 1,600 train / 400 test, 13 features). We address **Regression** (predicting addiction score, 0–10) and **Classification** (predicting addiction level $\in \{\text{Low, Medium, High}\}$) using a **Multi-Layer Perceptron** (MLP) and **K-Means** clustering extended to classification.

2 Methodology

2.1 Data Preparation

Training samples are **randomly permuted** then split into **80% train** (1,280) and **20% validation** (320), with the 400 test samples reserved for final evaluation. All features are **z-score normalised** ($z = (x - \mu)/\sigma$) with μ, σ computed on training data only to prevent data leakage.

2.2 Multi-Layer Perceptron

The MLP has a single hidden layer of dimension h with **ReLU** activation, and weights initialised using **Xavier/Glorot** uniform initialisation. The output layer uses **Sigmoid** for classification (3 outputs) and **Identity** for regression (1 output). Training uses **mini-batch SGD** with the dataset reshuffled at each epoch.

Loss functions. Classification uses **Cross-Entropy** and regression uses **MSE**. Our back-propagation applies the chain rule element-wise: $\delta^{(l)} = (\delta^{(l+1)} W^{(l+1)\top}) \odot \sigma'(a^{(l)})$, with weight update $W^{(l)} \leftarrow W^{(l)} - \eta z^{(l-1)\top} \delta^{(l)}$. Sigmoid is therefore used rather than Softmax: Softmax’s Jacobian is a full matrix, so element-wise application yields the wrong gradient unless CE and Softmax are analytically fused. Sigmoid’s derivative is genuinely diagonal, making it fully compatible with this implementation.

2.3 K-Means

Standard K-Means runs for up to 300 iterations with random initialisation, stopping when centres converge (`np.allclose`). Empty clusters are re-initialised to a random training point. After fitting, each cluster is assigned the **majority label** among its training members. Unseen points are classified by their nearest cluster’s label.

3 Experiments & Results

3.1 Hyperparameter Tuning

Hyperparameters were selected on the validation set. Figure 1 shows the K-Means and MLP classification results.

K-Means. $K \leq 4$ produces the same accuracy as the dummy classifier (69.69%), as too few clusters cannot capture intra-class structure. Accuracy peaks at $K=5$ (78.4%) but F1 keeps growing, reaching its maximum at $K=20$. Due to sensitivity to random initialisation, K was selected based on mean performance over **10 independent seeds** (mean F1 = 0.560 ± 0.035 at $K=20$), confirming the result is robust.

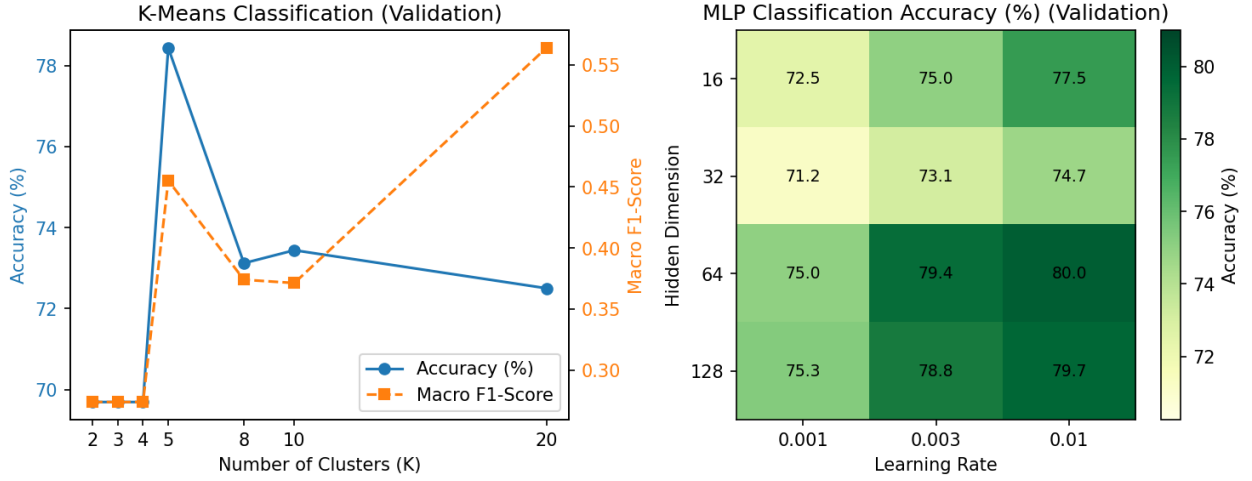


Figure 1: Validation performance. *Left*: K-Means accuracy and Macro F1 vs. K . *Right*: MLP classification accuracy (%) heatmap over hidden dimension and learning rate (300 epochs, batch size 32).

MLP Classification. Hidden dimension $h=64$ with $lr=0.01$ achieves the best validation accuracy (80.0%). Larger architectures ($h=128$) yield slightly lower performance, suggesting the dataset is too small to benefit from additional capacity.

MLP Regression. $lr=0.001$ consistently yields the lowest MSE across all architectures. Higher learning rates cause overfitting, especially for larger networks ($h=128$, $lr=0.01$: $MSE=1.42$). We select $h=32$, $lr=0.001$.

3.2 Final Test Set Evaluation

All models were retrained on the full 1,600 training samples and evaluated on the 400 test samples. Results are reported in Table 1.

Table 1: Final test set performance.

Method	HP	Acc.	F1	MSE
Dummy	–	70.00%	0.275	–
K-Means	$K=20$	74.25%	0.560	–
MLP	$h=64$, $lr=0.01$	80.00%	0.472	–
MLP (reg.)	$h=32$, $lr=0.001$	–	–	1.028

4 Discussion & Conclusion

Classification. The MLP achieves the best accuracy (80.00%) while K-Means leads in Macro F1 (0.560 vs. 0.472). This divergence is explained by severe class imbalance (Low: 280 samples, Medium: 107, High: 13). The MLP achieves near-perfect recall on Low (98.6%) but completely fails on High (recall = 0.0%), predicting it as Low or Medium. K-Means ($K=20$) recovers the High class well (recall = 84.6%) at the cost of Medium (7.5%). Both methods substantially outperform the dummy baseline ($F1=0.275$).

Regression. The MLP achieves $MSE=1.028$. Notably, the closed-form Linear Regression from Milestone 1 achieved $MSE=0.994$, outperforming the MLP. This suggests the addiction score has a near-linear relationship with the features that a neural network does not exploit more effectively on this small dataset.

Comparison with Milestone 1. Logistic Regression (M1) reached 88.25% accuracy and $F1=0.819$, substantially outperforming the MLP (80.00%, $F1=0.472$). This confirms the known tendency of neural networks to require larger datasets to surpass well-tuned parametric methods.

Conclusion. On this small, severely imbalanced dataset, simpler parametric methods remain competitive or superior. K-Means with many clusters is a strong classifier for minority classes; its low seed variance ($F1 \pm 0.035$ at $K=20$) confirms the result is robust. The MLP is flexible but struggles with rare classes and would benefit from class-weighting, regularisation, or more data.